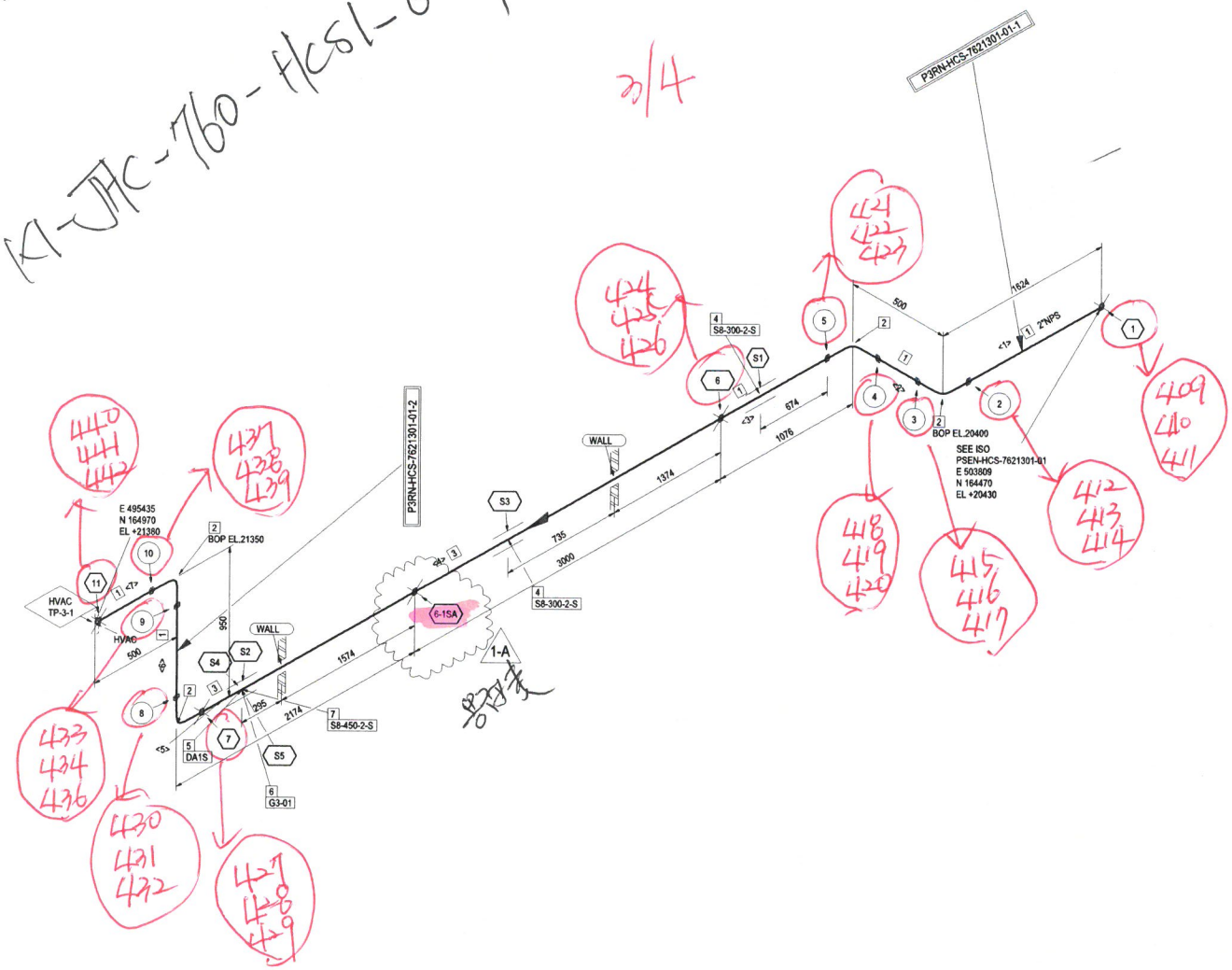
K1-JTC-760-HCSL-007

3/4



BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2"	4.2 M		P	PA02-000001902
2	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	2"	4		E	PB99-000038409

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
3	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2"	5.1 M		P	PA02-000001902
4	S8-300-2-S	2"	2			S8
5	DA1S	2"	1			DA1S
6	G3-01	2"	1			G3
7	S8-450-2-S	2"	1			S8

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	1548	2		<2>	348	2	
<3>	1000	2		<4>	3000	2	
<5>	2096	2		<6>	798	2	
<7>	414	2					

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2026-01-30

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRFSS. (bar.g)	OPER. TEMP. (℃)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (℃)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
2	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
1-A	30-01-26		J.Y.KIM	M.H.SUNG	T.H.LIM
1	24-09-25	ISSUED FOR REVISION	J.Y.KIM	M.H.SUNG	T.H.LIM
0	24-06-25	ISSUED FOR INFORMATION	J.Y.KIM	M.H.SUNG	T.H.LIM

PROJECT: PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P3RN
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480	FLUID	HCS
GAD NO.	-			LINE NO.	2-HCS-7621301-S98-IHI		
DWG NO.	P3RN-HCS-7621301-01			SH'T NO.	1 OF 1	REV.	1-A

PIPE modified at 9:30:25 Aug 2025 , Drawing Created at 15:56:57 25 August 2025
PIPE NAME : 2-HCS-7621301-S98-IHI_P3RN-01

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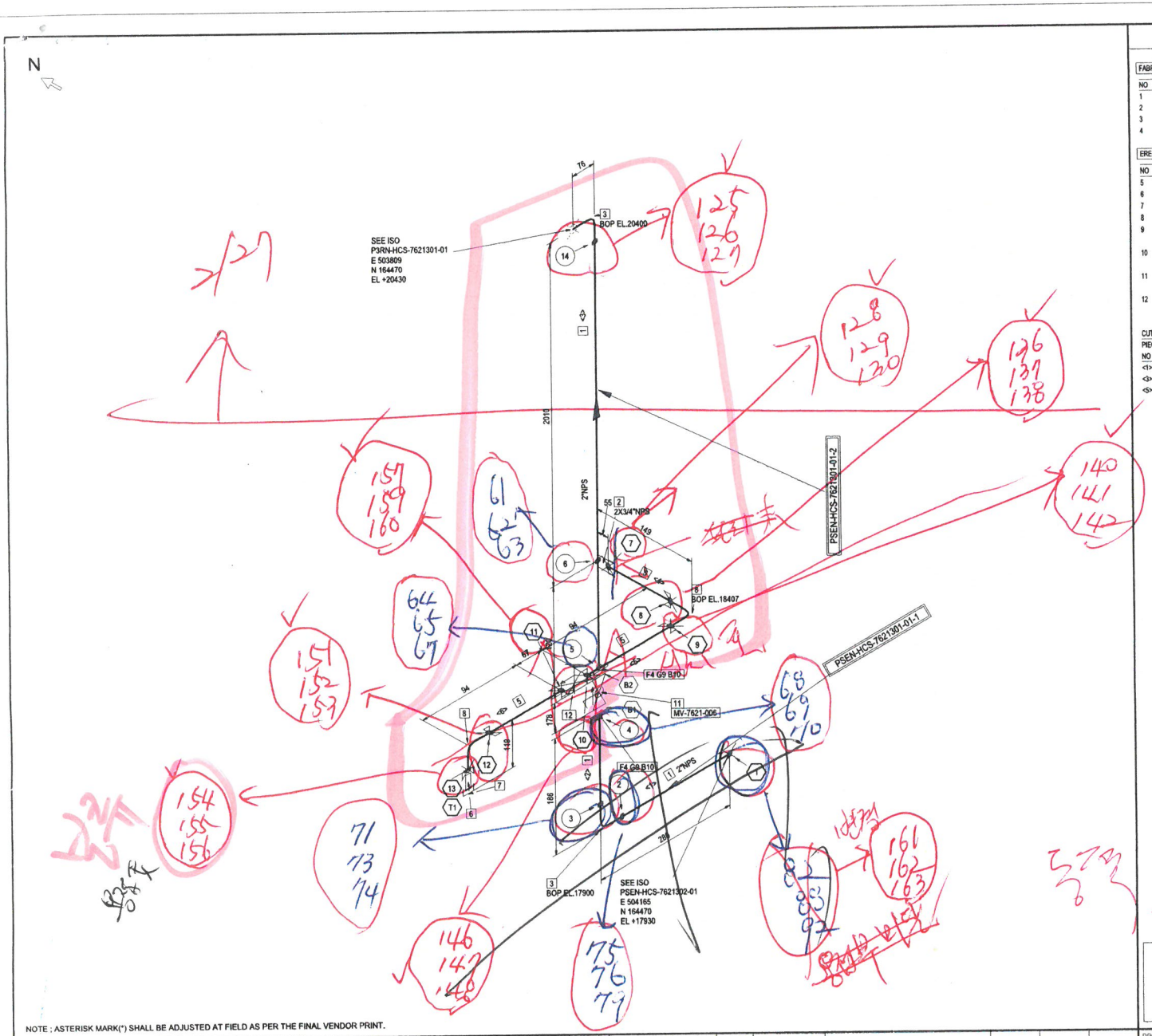

BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2"	2.4 M	S98	P	PA02-000001902
2	HALF COUPLING SW CL3000 ASTM A182-F316/316L	2X3/4"	1	S98	TH	PB99-000038951
3	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	2"	2	S98	E	PB99-000038409
4	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	2"	2	S98	F	PB01-000023955

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
5	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	0.3 M	S98	P	PA02-000000956
6	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
7	CAP SCR'D (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
8	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	2	S98	E	PB99-000038441
9	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHITE 4.5MM(0.18") THK.	2"	2	S98	G	PF01-000009267
10	90 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UN/UNR x MM	5/8"	8	S98	B	PB99-000076892
11	BALL VALVE FLGD-RF CL150 ASTM A351-CF8M SS316 TRIM PTFE SEAT FLOATING BALL FB LV. OP. (VAB160F)	2"	1	S98	VL	PP04-000007540
12	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	204	2		<2>	100	2	
<3>	2044	2		<4>	75	3/4	
<5>	75	3/4		<6>	75	3/4	

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-07-25

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
2	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
0	25-07-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	PSEN
P&ID NO	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480	FLUID	HCS
CAD NO.	-			LINE NO.	2-HCS-7621301-S98-IHI		
DWG NO.	PSEN-HCS-7621301-01					SH'T NO. 1 OF 1	REV 0

PIPE modified at 10:58:11 Jul 2025 Drawing Created at 10:07:47 18 July 2025
PIPE NAME : 2-HCS-7621301-S98-IHI_PSEN-01

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